Data Quality Strategy 2025-2027

Company A - Saudi Arabian Government Healthcare Entity

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1.0	March 2024	Ahmed Al-Rashid	Initial draft strategy
1.1	June 2024	Data Quality Team	Added Informatica IDQ architecture details
1.2	August 2024	Ahmed Al-Rashid	Updated KPI targets based on Q2 results
1.3	October 2024	Layla Al-Harbi	Added PDPL compliance requirements
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Approval Signatures

Role	Name	Signature	Date

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1. Executive Summary

1.1 Strategy Overview

This Data Quality Strategy defines Company A's comprehensive approach to achieving and maintaining high-quality, trusted data across all systems and processes for the period 2025-2027. The strategy is built on the foundation of "Zero Bad Data" in Critical Data Elements (CDEs) and supports our compliance with NDMO Domain 3 (Data Quality) requirements.

Key Objectives:
- Achieve 95% Data Trust Score (DTS) by end of 2025 (up from 92.4% in 2024)
- Cover 100% of Critical Data Elements with automated quality monitoring by 2026
- Reduce data quality issues by 60% through proactive prevention and automated remediation
- Establish a sustainable data quality culture with clear accountability and ownership

1.2 Strategic Pillars

Pillar 1: PEOPLE - Build a culture of data quality with clear roles, accountabilities, and training

Pillar 2: PROCESS - Implement standardized processes for data quality issue detection, remediation, and prevention

Pillar 3: TECHNOLOGY - Deploy Informatica Data Quality (IDQ) platform for automated monitoring and remediation

Pillar 4: METRICS - Establish transparent, actionable metrics that drive continuous improvement

1.3 Business Impact

Current State (2024):
- 92.4% Data Trust Score across 8 critical systems
- SAR 2.5 million annual cost due to poor data quality (rework, delayed decisions, compliance penalties)
- 50% of Critical Data Elements covered by automated quality checks
- 23,750 active data quality issues requiring manual remediation

Target State (2027):
- 98% Data Trust Score across all systems
- SAR 1.0 million annual cost (60% reduction through prevention and automation)
- 100% of Critical Data Elements covered by automated quality checks
- 40% auto-fix rate for data quality issues (minimal manual intervention)

Return on Investment (ROI):

- Investment: SAR 1.8 million over 3 years (technology, staff, training)
- Annual Savings: SAR 1.5 million (reduced rework, faster decision-making, avoided penalties)
- 3-Year ROI: 250% (SAR 4.5M savings vs SAR 1.8M investment)

1.4 Alignment with Organizational Strategy

This Data Quality Strategy directly supports:

1. NDMO Compliance: Addresses all 28 controls in NDMO Domain 3 (Data Quality)
2. PDPL Compliance: Ensures personal data accuracy and completeness (PDPL Article 7)
3. Digital Transformation: Enables AI/ML initiatives and advanced analytics with trusted data
4. Operational Excellence: Reduces manual rework and accelerates decision-making
5. Patient Safety: Ensures clinical data accuracy for safe, effective patient care

2. Strategic Vision

2.1 The "Zero Bad Data" Vision

Vision Statement:

> "By 2027, Company A will operate with Zero Bad Data in all Critical Data Elements, enabling trusted decision-making, seamless operations, and exceptional patient care. Our data will be so reliable that clinical staff, executives, and regulators can confidently act on it without hesitation."

What "Zero Bad Data" Means:

"Zero Bad Data" does NOT mean:
- ■ 100% perfection across every single data field in every system
- ■ Zero tolerance for any data quality issue
- ■ Elimination of all data entry errors

"Zero Bad Data" DOES mean:
- ■ 100% quality in Critical Data Elements (CDEs) used for regulatory reporting and critical operations
- ■ Proactive prevention of data quality issues at the point of entry
- ■ Automated detection and remediation of issues within 24 hours
- ■ Continuous monitoring with real-time alerts and dashboards

2.2 Business Case for Data Quality

2.2.1 The Cost of Poor Data Quality

Company A conducted a comprehensive assessment of data quality costs in 2024:

Cost Category	Annual Cost (SAR)	Examples
Rework and Manual Corrections	1,200,000	Staff time spent fixing errors, re-entering data

Delayed Decisions	580,000	Executives delaying strategic decisions due to unt
Compliance Penalties	320,000	NDMO audit findings, PDPL violation warnings
Lost Revenue	180,000	Incorrect billing, denied insurance claims
Patient Safety Incidents	150,000	Near-misses due to incorrect medication data
Reputational Damage	70,000	Patient complaints, negative social media
TOTAL	SAR 2,500,000	—

Key Findings:
- 68% of data quality issues are caused by manual data entry errors
- 52% of data quality issues could be prevented with real-time validation at point of entry
- 23% of staff time in Finance and HR departments is spent fixing data errors
- 18,500 patient records have at least one data quality issue (23% of patient base)

2.2.2 The Value of High-Quality Data

High-quality data enables:

1. Faster Decision-Making: Executives trust reports and act immediately (estimated 30% faster decisions)
2. Regulatory Compliance: Pass NDMO and PDPL audits with zero findings (avoid SAR 500K+ penalties)
3. Operational Efficiency: Reduce rework by 60% (free up 120 FTE hours/month for value-added work)
4. AI/ML Enablement: Launch predictive analytics and AI initiatives (requires 95%+ data quality)
5. Patient Safety: Prevent medication errors and treatment delays (zero tolerance for errors in clinical data)

2.3 Alignment with NDMO Domain 3

NDMO Domain 3: Data Quality consists of 28 controls across 4 sub-domains:

NDMO Control	Requirement	Company A Implementation
3.1.1 - Data Quality Framework	Establish DQ dimensions and standards	■ 6-dimension DAMA framework (Section 4.1)
3.1.2 - Critical Data Elements	Identify and prioritize CDEs	■ 180 CDEs identified (Appendix A)
3.2.1 - Data Profiling	Conduct regular data profiling	■ Monthly profiling via Informatica IDQ
3.2.2 - Data Quality Rules	Define and enforce DQ rules	■ 100+ rules in Rules Library (Appendix B)
3.3.1 - Issue Remediation	Track and remediate DQ issues	■ ServiceNow workflow (Section 6.2)
3.4.1 - DQ Metrics	Monitor and report DQ metrics	■ Data Trust Score dashboard (Section 9.1)

Compliance Status: Company A is 92.4% compliant with NDMO Domain 3 as of Q4 2024. This strategy targets 100% compliance by Q2 2025.

3. Current State Assessment

3.1 Data Quality Baseline (2024)

Overall Data Trust Score (DTS): 92.4% (as of November 2024)

System-Level Data Quality Scores:

System	DTS	Completeness	Accuracy	Timeliness	# Issues	Trend
Epic EHR	96.2%	98.1%	95.4%	94.8%	4,200	↑ +2.1%
Oracle EBS (Finance)	94.8%	97.2%	93.5%	93.2%	2,800	↑ +1.8%
SAP HR	91.5%	94.8%	89.2%	90.5%	1,250	↑ +3.2%
Salesforce CRM	88.3%	92.1%	85.2%	87.6%	8,500	↑ +1.5%
Philips PACS	97.8%	99.2%	97.1%	97.2%	450	■ Stable
Laboratory System	95.5%	98.5%	94.2%	93.8%	1,850	↑ +2.4%
Pharmacy System	96.8%	99.1%	95.8%	95.5%	1,200	↑ +1.9%
Data Warehouse	89.2%	93.5%	86.4%	87.8%	3,500	↑ +4.1%
OVERALL	92.4%	96.5%	89.6%	90.1%	23,750	↑ +2.2%

Key Insights:
- Salesforce CRM has the lowest quality (88.3%) - primarily due to incomplete lead/contact data
- Philips PACS has the highest quality (97.8%) - medical imaging metadata is highly structured
- Accuracy is the weakest dimension (89.6%) - many invalid email addresses, phone numbers, National IDs

3.2 Key Data Quality Issues

Top 10 Data Quality Issues (by Volume):

Rank	Issue	Count	Systems Affected	Business Impact
1	Invalid email addresses	8,250	Salesforce CRM, Epic EHR	Cannot send appointment reminders
2	Duplicate patient records	3,100	Epic EHR	Fragmented patient history
3	Incomplete National ID numbers	2,850	Epic EHR, SAP HR	PDPL compliance violation
4	Inactive mobile phone numbers	2,400	Epic EHR, Salesforce	Cannot contact patients
5	Incorrect billing codes	1,950	Oracle EBS	Denied insurance claims
6	Null values in mandatory fields	1,800	Multiple systems	Reporting errors
7	Out-of-range dates (default: 1900-01-01)	1,200	SAP HR, Oracle EBS	Incorrect calculations
8	Inconsistent department names	950	SAP HR, Epic EHR	Reporting fragmentation
9	Trailing spaces in text fields	720	Multiple systems	Data matching failures
10	Invalid IBAN formats	530	Oracle EBS	Failed vendor payments
TOTAL	—	23,750	—	—

3.3 Root Cause Analysis

Why do data quality issues occur?

Root Cause	% of Issues	Examples	Prevention Strategy
Manual data entry errors	68%	Typos, copy-paste errors	Real-time validation at point of entry
Lack of validation rules	18%	No email format validation	Implement 100+ DQ rules in IDQ
System integrations	8%	Data lost during ETL	Add DQ checks in data pipelines
Legacy data migration	4%	Inherited bad data from old systems	One-time cleanup project
Outdated data	2%	Patients change phone numbers	Periodic data refresh campaigns

Key Finding: 68% of issues are preventable with real-time validation at point of entry. This is the focus of our 2025 implementation.

4. Data Quality Framework

4.1 The 6 Dimensions of Data Quality (DAMA)

Company A adopts the DAMA International 6-dimension framework:

Dimension 1: Completeness

Definition: Are all required data fields populated?

Measurement:
Completeness % = (# of non-null values / # of mandatory fields) × 100

Example Rule:
- `NATIONAL_ID` must be present for all Saudi employees (SAP HR)
- `PATIENT_NAME` must be present for all patient records (Epic EHR)

Target: 100% completeness for all Critical Data Elements (CDEs)

Dimension 2: Accuracy

Definition: Does the data correctly reflect the real-world object?

Measurement:
- Sample-based verification (e.g., call 100 random phone numbers to verify accuracy)
- Automated validation (e.g., email bounce rate, IBAN checksum validation)

Example Rule:

- `PHONE_NUMBER` must be verified via SMS OTP for new patient registrations
- `IBAN` must pass mod-97 checksum validation

Target: 98% accuracy for CDEs (allowing 2% for edge cases and exceptions)

Dimension 3: Consistency

Definition: Is the same data represented the same way across systems?

Measurement:
Consistency % = (# of matching values across systems / # of expected matches) × 100

Example Rule:
- `EMPLOYEE_ID` in SAP HR must match `USER_ID` in Epic EHR for all clinical staff
- `DEPARTMENT_NAME` must use standardized values from master data list

Target: 95% consistency across integrated systems

Dimension 4: Timeliness

Definition: Is the data available when needed and up-to-date?

Measurement:
- Data freshness: time elapsed since last update
- SLA compliance: % of data delivered within agreed timeframe

Example Rule:
- Patient demographic changes (address, phone) must be updated within 24 hours of notification
- Financial reports must be generated within 3 business days of month-end

Target: 95% of data updated within defined timeliness SLAs

Dimension 5: Validity

Definition: Does the data conform to defined formats and business rules?

Measurement:
Validity % = (# of records passing validation rules / total # of records) × 100

Example Rule:
- `EMAIL` must match regex pattern: `^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$`
- `NATIONAL_ID` for Saudi nationals must be exactly 10 digits starting with `1` or `2`

Target: 100% validity for CDEs (enforced via database constraints and API validation)

Dimension 6: Uniqueness

Definition: Is each real-world entity represented only once in the data?

Measurement:
Uniqueness % = (# of unique records / total # of records) × 100

Example Rule:
- Each patient must have exactly 1 active record in Epic EHR (no duplicates)
- Each vendor must have exactly 1 record in Oracle EBS

Target: 100% uniqueness for master data entities (patients, employees, vendors)

4.2 Critical Data Elements (CDEs)

What are Critical Data Elements?

CDEs are data fields that are essential for business operations, regulatory compliance, or decision-making. They receive the highest priority for data quality monitoring and remediation.

CDE Selection Criteria:

A data element is classified as "Critical" if it meets any of the following criteria:
1. Regulatory Requirement: Required for NDMO, PDPL, or Ministry of Health reporting
2. Patient Safety: Used in clinical decision-making or medication administration
3. Financial Impact: Used for billing, payments, or financial reporting
4. Executive Reporting: Appears in CEO/Board dashboards or KPIs

CDE Inventory Summary:

CDE Category	# of CDEs	Example Fields
Patient Demographics	45	National ID, Name, DOB, Gender, Phone, Email
Clinical Data	58	Diagnosis codes (ICD-10), medication orders, lab r
Financial Data	32	Invoice amounts, payment status, IBAN, billing cod
Employee Data	28	Employee ID, National ID, Department, Job Title
Vendor/Supplier Data	17	Vendor name, IBAN, tax registration number
TOTAL	180	—

Full CDE Inventory: See Appendix A for complete list of 180 Critical Data Elements.

4.3 Data Quality Rules

Data Quality Rules Library:

Company A has defined 100+ data quality rules across the 6 dimensions. These rules are implemented in Informatica Data Quality (IDQ) and enforced at:
1. Point of Entry: Real-time validation in web forms and mobile apps

2. Batch Processing: Nightly scans of databases to detect issues
3. Data Integration: ETL pipelines validate data before loading

Example Rules (Top 10):

Rule ID	Rule Name	Dimension	Logic	Priority
DQR-001	National ID Format Validation	Validity	National ID must be 10 digits (Saudi) or valid for	CRITICAL
DQR-002	Email Format Validation	Validity	Email must match regex: ^[a-zA-Z0-9._%+-]+@[a-zA-Z	HIGH
DQR-003	Phone Number Verification	Accuracy	Phone must be verified via SMS OTP or call-back	HIGH
DQR-004	Duplicate Patient Detection	Uniqueness	No two patients with same National ID + DOB + Name	CRITICAL
DQR-005	Mandatory Field Completeness	Completeness	12 mandatory fields must be non-null for patient r	CRITICAL
DQR-006	IBAN Checksum Validation	Validity	IBAN must pass mod-97 checksum	HIGH
DQR-007	Date Range Validation	Validity	DOB must be between 1900-01-01 and today	HIGH
DQR-008	Department Name Consistency	Consistency	Department names must match master data list (45 v	MEDIUM
DQR-009	Data Freshness Check	Timeliness	Patient contact info updated within last 12 months	MEDIUM
DQR-010	Trailing Space Detection	Validity	Text fields must not have leading/trailing spaces	LOW

Full Rules Library: See Appendix B for all 100+ data quality rules.

5. Technology Architecture

5.1 Informatica Data Quality (IDQ) Platform

Tool Selection Rationale:

Company A selected Informatica Data Quality (IDQ) as the enterprise data quality platform based on:
1. Integration with Informatica EDC (Data Catalog): Seamless integration for metadata-driven DQ rules
2. Gartner Magic Quadrant Leader: Top-ranked DQ tool for 8 consecutive years
3. Saudi Arabia Deployment: Can be deployed in Saudi cloud (AWS me-south-1 Bahrain)
4. Scalability: Handles 18.5M patient records + 12 TB data warehouse

IDQ Deployment Architecture:

■■■
■ INFORMATICA IDQ PLATFORM ■
■■■
■ ■
■ ■■■■■■■■■■■■ ■■■■■■■■■■■■ ■■■■■■■■■■■■ ■
■ ■ IDQ Data ■ ■ IDQ Analyst ■ ■ IDQ Developer■ ■
■ ■ Director ■ ■ ■ ■ Tool ■ ■
■ ■ (Monitoring) ■ ■ (Profiling) ■ ■ (Rule Design)■ ■

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■ ■ IDQ ENGINE (Processing & Validation) ■ ■
■ ■ - Batch Processing (Nightly Jobs) ■ ■
■ ■ - Real-Time API (Point-of-Entry Validation) ■ ■
■ ■ - Data Profiling & Discovery ■ ■
■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■
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■ DATA SOURCES ■ ■ ISSUE TRACKING ■
■■■■■■■■■■■■■■■■■■■■ ■■■■■■■■■■■■■■■■■■■■
■ - Epic EHR ■ ■ - ServiceNow ■
■ - Oracle EBS ■ ■ (Tickets) ■
■ - SAP HR ■ ■ - Power BI ■
■ - Salesforce CRM ■ ■ (Dashboards) ■
■ - Data Warehouse ■ ■ - Email Alerts ■
■■■■■■■■■■■■■■■■■■■■ ■■■■■■■■■■■■■■■■■■■■
```

Key IDQ Components:

Component	Purpose	Users
IDQ Data Director	Monitor data quality dashboards, view scorecards	CDO, Data Stewards, Executives
IDQ Analyst	Perform data profiling, analyze data quality trend	Data Quality Analysts
IDQ Developer	Design and configure data quality rules, mappings	Data Quality Engineers
IDQ Engine	Execute DQ rules, validate data, generate alerts	Automated (no direct users)

5.2 Deployment Patterns (Batch & Real-Time)

Company A uses two deployment patterns for data quality monitoring:

Pattern 1: Batch Processing (Nightly Jobs)

Use Case: Scan large databases (Epic EHR, Oracle EBS, Data Warehouse) to detect existing data quality issues

Schedule: Daily at 2:00 AM (off-peak hours)

Process Flow:
1. 2:00 AM: IDQ connects to source databases (read-only access)

2. 2:05 AM: IDQ runs 100+ data quality rules against all CDEs
3. 3:30 AM: IDQ generates data quality scorecard (system-level, dimension-level, rule-level)
4. 3:45 AM: IDQ creates ServiceNow tickets for new issues (assigned to Data Stewards)
5. 4:00 AM: IDQ sends email summary to CDO and Data Quality Team
6. 8:00 AM: Power BI dashboard refreshes with latest scores

Output:
- Data Quality Scorecard: System-level scores for each of 8 systems
- Issue List: 23,750 active issues with severity, owner, due date
- ServiceNow Tickets: Auto-created tickets for new issues

Pattern 2: Real-Time API (Point-of-Entry Validation)

Use Case: Prevent bad data from entering systems by validating at the point of data entry

Integration: IDQ exposes REST API called by front-end applications (web forms, mobile apps)

Process Flow:
1. User Action: User fills out patient registration form and clicks "Save"
2. API Call: Front-end app calls POST /api/idq/validate with patient data
3. IDQ Validation: IDQ runs 12 validation rules on patient data (National ID format, email format, phone format, etc.)
4. Response: IDQ returns validation result (PASS or FAIL with error messages)
5. User Feedback: If FAIL, form displays error messages; user corrects and resubmits
6. Save: If PASS, data is saved to Epic EHR

Example API Request:

json

```
POST /api/idq/validate
{
"entity_type": "PATIENT",
"data": {
"national_id": "1234567890",
"name": "Ahmed Al-Rashid",
"dob": "1985-03-15",
"phone": "+966501234567",
"email": "ahmed@example.com"
}
}
```

Example API Response (FAIL):

json

```
{
"status": "FAIL",
"errors": [
```

{
"rule_id": "DQR-002",
"field": "email",
"message": "Email domain 'example.com' is not valid. Please use a real email address."
},
{
"rule_id": "DQR-003",
"field": "phone",
"message": "Phone number has not been verified. Please click 'Send OTP' to verify."
}
]
}

Benefits:
- Prevention over Detection: Stop bad data before it enters the system
- Immediate User Feedback: Users correct errors in real-time (better user experience)
- Reduced Remediation Backlog: Fewer issues to fix after the fact

Target: By end of 2025, 80% of data entry forms will integrate with IDQ real-time API.

5.3 Integration with Data Catalog

Metadata-Driven Data Quality:

Informatica IDQ is tightly integrated with Informatica Enterprise Data Catalog (EDC) to enable metadata-driven data quality:

1. CDE Tagging: All 180 Critical Data Elements are tagged as "CDE" in the Data Catalog
2. Rule Assignment: Data Quality rules are linked to specific data elements in the Catalog
3. Steward Assignment: Each data element has a designated Data Steward (auto-populated in DQ tickets)
4. Automated Discovery: When IDQ detects an issue, it looks up the Catalog to find the Steward and system

Example:
- IDQ detects invalid email in `Epic EHR.PATIENT_CONTACT.EMAIL` field
- IDQ queries Catalog API: "Who is the Steward for `Epic EHR.PATIENT_CONTACT.EMAIL`?"
- Catalog responds: "Steward = Layla Al-Harbi (Patient Services Manager)"
- IDQ creates ServiceNow ticket assigned to Layla Al-Harbi

6. Operating Model

6.1 Roles and Responsibilities

Role	Responsibilities	Count	Time Commitment

Chief Data Officer (CDO)	Executive sponsor; approve DQ strategy; quarterly	1	5% (2 hrs/week)
Data Quality Manager	Lead DQ program; manage team; report to CDO	1	100% (FTE)
Data Quality Analysts (DQAs)	Configure IDQ rules; analyze trends; support Stewa	3	100% (FTE each)
Data Stewards	Remediate DQ issues in their domain; approve excep	24	10-20% (varies)
System Owners	Provide read/write access to systems; approve chan	8	5%
Data Governance Committee	Approve DQ policies; escalations; quarterly oversi	12	2 hrs/quarter

Data Steward Coverage:

System	Data Steward	Department	# of CDEs
Epic EHR	Layla Al-Harbi	Patient Services	103
Oracle EBS	Noura Al-Shahrani	Finance	32
SAP HR	Abdullah Al-Mansour	Human Resources	28
Salesforce CRM	Fatima Al-Zahrani	Marketing	17

6.2 Data Quality Issue Remediation Workflow

The "Fix-It Workflow":

■■■■■■■■■■■■■■■■■
■ 1. DETECT ■ IDQ nightly job detects invalid email (DQR-002 violation)
■■■■■■■■■■■■■■■■■
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■■■■■■■■■■■■■■■■■
■ 2. ROUTE ■ IDQ queries Catalog: Who is Steward for this field?
■■■■■■■■■■■■■■■■■ → Steward = Layla Al-Harbi (Patient Services Manager)
↓
■■■■■■■■■■■■■■■■■
■ 3. TICKET ■ IDQ creates ServiceNow ticket:
■■■■■■■■■■■■■■■■■ - Assigned to: Layla Al-Harbi
■ - Priority: HIGH
■ - Due date: 5 business days
■ - Description: "Patient record #12345 has invalid email"
↓
■■■■■■■■■■■■■■■■■
■ 4. FIX ■ Layla logs into Epic EHR and corrects patient email
■■■■■■■■■■■■■■■■■ - Old value: "test@@gmail..com" (invalid)
■ - New value: "ahmed@gmail.com" (valid)
↓
■■■■■■■■■■■■■■■■■
■ 5. RE-RUN ■ Next nightly job (2:00 AM) re-validates record

■■■■■■■■■■■■■■■■ → Rule DQR-002 now PASSES
↓
■■■■■■■■■■■■■■■■
■ 6. CLOSE ■ IDQ auto-closes ServiceNow ticket
■■■■■■■■■■■■■■■■ - Status: RESOLVED
- Resolution: "Email corrected by Data Steward"

Service Level Agreements (SLAs):

Issue Priority	SLA (Resolution Time)	Auto-Escalation
CRITICAL (patient safety, regulatory)	1 business day	Escalate to CDO after 4 hours
HIGH (financial, operational)	5 business days	Escalate to Manager after 3 days
MEDIUM (reporting, analytics)	15 business days	Escalate to Manager after 10 days
LOW (cosmetic, convenience)	30 business days	No escalation

6.3 Exception Handling Process

What is an Exception?

An exception is a data quality issue that appears to violate a rule, but is actually legitimate due to unique business circumstances.

Example Exceptions:
- VIP patient with special National ID format (issued by Royal Court)
- Historical data from legacy system with different format
- Test/Training records intentionally using fake data

Exception Approval Process:

1. Steward Reviews Issue: Steward determines issue is a legitimate exception (not an error)
2. Steward Requests Exception: Steward clicks "Request Exception" button in ServiceNow ticket
3. Steward Provides Justification: Steward documents why this is an exception (business reason, approval)
4. DQ Manager Approves: Data Quality Manager reviews and approves/rejects exception
5. IDQ Whitelists Record: If approved, IDQ adds record to exception whitelist (will not flag in future scans)
6. Quarterly Audit: DQ Manager reviews all exceptions quarterly to ensure they are still valid

Exception Governance:
- Maximum 2% exception rate per rule (if >2%, rule needs to be redesigned)
- Quarterly review of all exceptions by Data Quality Manager
- Annual purge of expired exceptions (e.g., test records deleted, historical data archived)

7. Data Governance Integration

7.1 Data Stewardship Model

Data Quality is a shared responsibility between:
- Data Quality Team: Define rules, configure tools, analyze trends
- Data Stewards: Own specific data domains, remediate issues, approve exceptions
- System Owners: Provide access, approve system changes, fund improvements

Steward Accountability:
- Each Data Steward has a monthly DQ target (e.g., "Resolve 90% of issues within SLA")
- Steward performance is reviewed quarterly by their manager and CDO
- Top-performing Stewards are recognized in company newsletter

7.2 Data Quality Council

Purpose: Governance body for escalations, policy approvals, and strategic direction

Membership: 12 members (CDO, Data Quality Manager, 8 System Owners, 2 Steward representatives)

Meeting Cadence: Quarterly (1 hour)

Agenda:
1. Review quarterly DQ scorecard and trends
2. Approve new DQ rules and policy changes
3. Review escalated issues and exceptions
4. Discuss strategic initiatives (e.g., AI/ML readiness)

7.3 Escalation Procedures

When to escalate:
- Issue cannot be resolved within SLA due to system limitations
- Steward disagrees with rule or believes rule is incorrect
- Cross-system issue requiring coordination between multiple Stewards

Escalation Path:
1. Level 1: Data Steward → Data Quality Manager (for technical issues, rule clarifications)
2. Level 2: Data Quality Manager → CDO (for policy decisions, resource requests)
3. Level 3: CDO → Data Quality Council (for strategic issues, major investments)

8. Implementation Roadmap

8.1 Phase 1: Foundation (Q1 2025)

Objective: Deploy IDQ platform, configure initial rules, train team

Milestone	Description	Due Date	Owner
M1.1	IDQ platform deployed in AWS (Bahrain region)	Jan 15, 2025	IT Infrastructure
M1.2	50 priority DQ rules configured in IDQ	Feb 1, 2025	DQ Analysts
M1.3	Integration with Epic EHR, Oracle EBS, SAP HR	Feb 15, 2025	DQ Engineers
M1.4	Data Steward training (2-day workshop)	Feb 20, 2025	DQ Manager
M1.5	Nightly batch processing operational	Mar 1, 2025	DQ Engineers
M1.6	Power BI dashboard launched	Mar 15, 2025	BI Team
M1.7	Phase 1 Go-Live	Mar 31, 2025	CDO

Success Criteria:
- IDQ processing 5 million records/day without errors
- 50 DQ rules operational across 3 systems
- All 24 Data Stewards trained and active in ServiceNow

8.2 Phase 2: Expansion (Q2-Q3 2025)

Objective: Expand coverage to all systems, deploy real-time API, achieve 95% DTS

Milestone	Description	Due Date	Owner
M2.1	Remaining 50 DQ rules configured (total: 100)	Apr 30, 2025	DQ Analysts
M2.2	Integration with Salesforce, PACS, Lab, Pharmacy	May 31, 2025	DQ Engineers
M2.3	Real-time API deployed (pilot: Epic patient regist	Jun 15, 2025	DQ Engineers
M2.4	Real-time API expanded to 10 forms	Jul 31, 2025	DQ Engineers
M2.5	Achieve 95% Data Trust Score	Sep 30, 2025	DQ Manager

Success Criteria:
- IDQ processing all 8 systems (18.5M records total)
- 100 DQ rules operational
- 95% Data Trust Score achieved
- 10 forms integrated with real-time API

8.3 Phase 3: Optimization (Q4 2025 - 2027)

Objective: Achieve 98% DTS, 40% auto-fix rate, 100% CDE coverage

Milestone	Description	Due Date	Owner
M3.1	Auto-remediation workflows for 20% of issues	Dec 31, 2025	DQ Engineers
M3.2	Real-time API integrated with all 50 data entry fo	Jun 30, 2026	DQ Engineers
M3.3	100% of CDEs covered by DQ rules	Dec 31, 2026	DQ Analysts

M3.4	Achieve 98% Data Trust Score	Jun 30, 2027	DQ Manager
M3.5	40% auto-fix rate for DQ issues	Dec 31, 2027	DQ Manager

Success Criteria:
- 98% Data Trust Score maintained for 6+ months
- 40% of issues auto-remediated without manual intervention
- Zero CRITICAL issues older than 1 day

9. Metrics, KPIs, and Targets

9.1 Data Trust Score (DTS)

Formula:

DTS = (Completeness × 0.30) + (Accuracy × 0.30) + (Uniqueness × 0.20) + (Validity × 0.10) + (Consistency × 0.05) + (Timeliness × 0.05)

Rationale for Weighting:
- Completeness and Accuracy are weighted highest (30% each) because they have the most direct impact on decision-making
- Uniqueness is weighted 20% because duplicates cause significant operational issues
- Validity, Consistency, Timeliness are weighted lower (5-10%) because they are important but less critical than completeness/accuracy

DTS Scoring Scale:

DTS Range	Rating	Interpretation
95% - 100%	■ EXCELLENT	Data is highly trusted; suitable for AI/ML and cri
90% - 94.9%	■ GOOD	Data is reliable for most use cases; minor improve
80% - 89.9%	■ FAIR	Data has notable issues; not suitable for critical
< 80%	■ POOR	Data quality is unacceptable; urgent remediation r

9.2 KPI Dashboard and Reporting

Monthly DQ Dashboard (Power BI):

The CDO Dashboard displays:
1. Overall DTS: Single score (0-100) with trend arrow (↑ ↓ ↔)
2. System-Level Scores: Table showing DTS for each of 8 systems
3. Dimension-Level Scores: Bar chart showing 6 dimension scores
4. Top 10 Issues: List of highest-impact DQ issues with assigned Stewards
5. Steward Performance: Table showing each Steward's issue resolution rate and SLA compliance

6. Trend Analysis: Line chart showing DTS over last 12 months

Audience:
- Executive Dashboard: CEO, CFO, COO (updated monthly)
- Operational Dashboard: Data Stewards, System Owners (updated daily)
- Technical Dashboard: Data Quality Team (updated hourly)

9.3 3-Year Targets (2025-2027)

Metric	2024 Actual	2025 Target	2026 Target	2027 Target
Data Trust Score (Overall)	92.4%	95%	97%	98%
Completeness	96.5%	98%	99%	99.5%
Accuracy	89.6%	94%	96%	98%
Uniqueness	94.2%	97%	99%	99.5%
Validity	91.8%	95%	97%	98%
Consistency	88.5%	92%	95%	96%
Timeliness	90.1%	93%	95%	96%
# of Active DQ Issues	23,750	15,000	8,000	5,000
CDEs Covered by DQ Rules	50% (90/180)	80% (144/180)	100% (180/180)	100%
Forms with Real-Time Validation	0%	20% (10/50)	60% (30/50)	100% (50/50)
Auto-Fix Rate	5%	20%	30%	40%
Steward SLA Compliance	78%	85%	90%	95%

10. Risks and Mitigation

Risk	Impact	Likelihood	Mitigation
Steward Resistance: Stewards view DQ as 'extra wor	HIGH	MEDIUM	- Include DQ targets in Steward performance review
Tool Performance: IDQ unable to process 18.5M reco	MEDIUM	LOW	- Conduct performance testing before go-live -
Budget Overruns: Implementation costs exceed SAR 1	HIGH	MEDIUM	- Phased rollout to spread costs over 3 years -
False Positives: DQ rules flag legitimate data as	MEDIUM	MEDIUM	- Pilot rules on sample data before production
Integration Failures: IDQ unable to connect to sou	HIGH	LOW	- Early integration testing in UAT environment

11. Budget and Resources

3-Year Budget (2025-2027):

Category	2025	2026	2027	Total
Software Licenses				
- Informatica IDQ (Enterprise)	SAR 480,000	SAR 480,000	SAR 480,000	SAR 1,440,000
- ServiceNow DQ Module	SAR 60,000	SAR 60,000	SAR 60,000	SAR 180,000
Cloud Infrastructure (AWS)	SAR 120,000	SAR 120,000	SAR 120,000	SAR 360,000
Personnel				
- Data Quality Manager (FTE)	SAR 240,000	SAR 250,000	SAR 260,000	SAR 750,000
- Data Quality Analysts (3 FTE)	SAR 480,000	SAR 500,000	SAR 520,000	SAR 1,500,000
- Data Steward Time (24 Stewards @ 10%)	SAR 180,000	SAR 180,000	SAR 180,000	SAR 540,000
Training & Change Management	SAR 80,000	SAR 40,000	SAR 40,000	SAR 160,000
Contingency (10%)	SAR 164,000	SAR 163,000	SAR 166,000	SAR 493,000
TOTAL	SAR 1,804,000	SAR 1,793,000	SAR 1,826,000	SAR 5,423,000

Funding Source: Approved in 2025-2027 IT Budget under "Data Management & Governance" portfolio.

12. Appendices

Appendix A: Critical Data Elements (CDE) Inventory

180 CDEs Across 5 Categories:

(Sample - full list in separate spreadsheet)

CDE ID	CDE Name	System	Table.Column	DQ Rules	Steward
CDE-001	Patient National ID	Epic EHR	PATIENT.NATIONAL_ID	DQR-001, DQR-005	Layla Al-Harbi
CDE-002	Patient Full Name	Epic EHR	PATIENT.FULL_NAME	DQR-005, DQR-008	Layla Al-Harbi
CDE-003	Patient Date of Birth	Epic EHR	PATIENT.DOB	DQR-005, DQR-007	Layla Al-Harbi
CDE-004	Patient Mobile Phone	Epic EHR	PATIENT_CONTACT.MOBILE	DQR-003, DQR-005	Layla Al-Harbi
CDE-005	Patient Email	Epic EHR	PATIENT_CONTACT.EMAIL	DQR-002, DQR-005	Layla Al-Harbi

Appendix B: Data Quality Rules Library (100 Rules)

(Sample - full list in Informatica IDQ system)

Rule ID	Rule Name	Dimension	SQL Logic	Severity
DQR-001	National ID Format Validation	Validity	LENGTH(NATIONAL_ID) = 10 AND REGEXP_LIKE(NATIONAL_	CRITICAL
DQR-002	Email Format Validation	Validity	REGEXP_LIKE(EMAIL, '^[A-Za-z0-9._%+-]+@[A-Za-z0-9.	HIGH
DQR-003	Phone Number Verification	Accuracy	PHONE_VERIFIED_FLAG = 'Y' OR PHONE_LAST_VERIFIED_D	HIGH
DQR-004	Duplicate Patient Detection	Uniqueness	COUNT() OVER (PARTITION BY NATIONAL_ID, DOB) = 1	CRITICAL
DQR-005	Mandatory Field Completeness	Completeness	NATIONAL_ID IS NOT NULL AND FULL_NAME IS NOT NULL	CRITICAL

Appendix C: Glossary of Terms

Term	Definition
CDE (Critical Data Element)	A data field essential for business operations, re
DTS (Data Trust Score)	Weighted aggregate score (0-100) measuring overall
Data Steward	Business user responsible for data quality in thei
IDQ (Informatica Data Quality)	Enterprise software platform for data profiling, v
False Positive	A data quality rule flagging legitimate data as an
Exception	A data quality issue that appears to violate a rul

END OF STRATEGY DOCUMENT

Document Approval:

Approver	Signature	Date
Dr. Khalid Al-Dosari, CEO	Digitally Signed	November 22, 2024
Ahmed Al-Rashid, CDO	Digitally Signed	November 15, 2024
Dr. Sara Al-Mutlaq, DGC Chair	Digitally Signed	November 22, 2024

Next Review Date: November 2025

This document supports Company A's compliance with NDMO Domain 3 (Data Quality) and PDPL Article 7 (Data Accuracy).

Company A - Trusted Data, Better Decisions, Exceptional Care